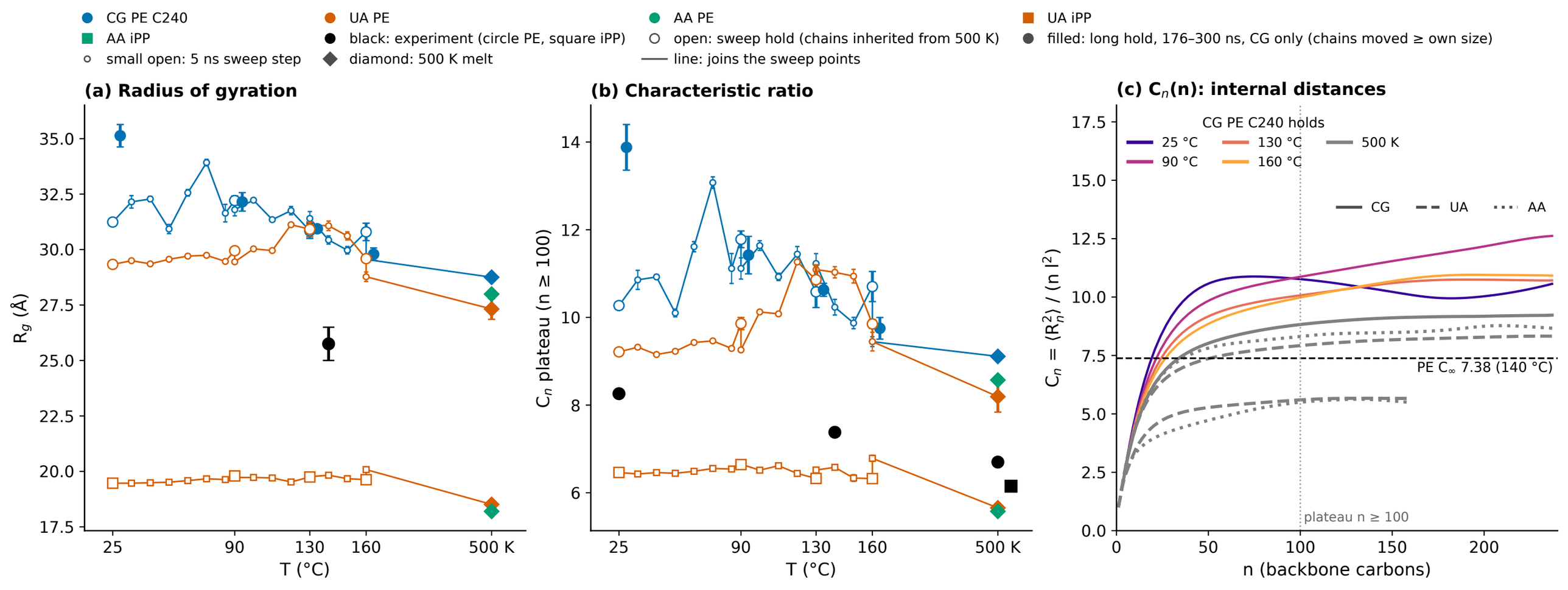

CG PE C240
AA iPP
small open: 5 ns sweep step
UA PE
black: experiment (circle PE, square iPP)
diamond: 500 K melt
AA PE
open: sweep hold (chains inherited from 500 K)
line: joins the sweep points
UA iPP
filled: long hold, 176–300 ns, CG only (chains moved ≥ own size)
(a) Radius of gyration
R_g (Å)
T (°C)
(b) Characteristic ratio
C_n plateau (n ≥ 100)
T (°C)
(c) C_n(n): internal distances
CG PE C240 holds
25 °C
90 °C
130 °C
160 °C
500 K
CG
UA
AA
PE C_∞ 7.38 (140 °C)
plateau n ≥ 100
$C_n = \langle R_n^2 \rangle / (n\,l^2)$
n (backbone carbons)